

SHRINIVAASAN L N

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PROFESSIONAL SUMMARY

AI-native Backend Engineer with 5+ years of experience in **high-scale microservices, API integrations, and full-stack development** at Backbase and TransUnion. Specialized in **legacy modernization, optimization for throughput and reliability**, and **GenAI/LLM agent integration** into enterprise workflows to drive substantial product value.

WORK EXPERIENCE

Senior Backend Engineer

Backbase

Mar 2026 – Present

Hyderabad, Telangana

- Owned backend engineering for the **Grand Central** domain, designing and building a **cloud-native iPaaS** to integrate heterogeneous Core Banking systems with Backbase's Engagement Banking Platform.
- Restored **100% reliability** across core banking flows by diagnosing and resolving a **critical production race condition** in a shared cryptographic pipeline under high concurrency.
- Restored system reliability by re-architecting a legacy static cache into a **thread-safe, CDI-managed singleton**, eliminating shared mutable state and TOCTOU vulnerabilities under concurrent load.
- Led **systematic performance optimization** across API middleware, reducing end-to-end latency by over **95%** (from approximately 1s to 50ms) through thread-pool tuning, JVM optimizations, and replacing Groovy-based transformations with JsonPath.
- Designed and implemented reusable **OAuth2 client configurators** for **HTTP** and **SOAP (Apache CXF)**, centralizing token acquisition, TLS certificate handling, and optional **mTLS** flows to standardize secure outbound authentication across connectors.
- Defined platform-wide **non-functional requirements** including **mTLS**, secure headers, and standardized error handling, embedding **security-by-design** across all integrations.

Backend Engineer

Backbase

May 2024 – Present

Hyderabad, Telangana

- Architected and owned a **high-throughput, event-driven** integration platform using **Azure Service Bus** to decouple Core Banking systems (JMS queues, webhooks) from downstream services, ensuring **reliable, fault-tolerant processing of business-critical banking events**.
- Automated the **Azure APIM** lifecycle using **Infrastructure-as-Code (YAML)** and optimized **GitHub Actions** CI/CD pipelines to ensure reproducible, version-controlled deployments.
- Built an **LLM-agnostic GenAI agent** using **Python** and **LangChain** to auto-generate complex **Jolt transformations** via a **RAG pipeline** backed by **ChromaDB**.
- Standardized **AI-assisted development workflows** at the organization level by configuring **GitHub Copilot** and **Cursor Bugbot**, and independently researched and developed custom agent skills and prompt specifications for tools such as **Antigravity** and **OpenCode** to optimize **model and tool selection** across different engineering workflows.
- Acted as **Scrum Master and technical liaison**, coordinating cross-team delivery across **6+ monthly releases** for banking and credit union clients; awarded **Q4 Excellence: Raising the Bar (2025)** for high-impact contributions.

Developer, Applications Development

TransUnion

Feb 2023 – Apr 2024

Chennai, Tamil Nadu

- Owned the **Batch Operator UI**, building and scaling complex **Angular (TypeScript)** applications used to monitor, pause, and control long-running batch and stream processing workflows.
- Solved frontend scalability constraints by implementing **server-side search, pagination, and virtual scrolling**, enabling responsive rendering of **100K+ row datasets** previously limited by network payload size.
- Maintained supporting **Spring Boot** backend services exposing batch execution state via **gRPC** and **REST**, supporting polling-based access patterns over **large batch metadata payloads** with consistency prioritized over latency.
- Led the migration from REST wrappers to **gRPC-Web** with **Envoy Proxy** to streamline browser-to-service communication and reduce middleware overhead.
- Performed database migrations and operational maintenance on **PostgreSQL**, including authoring migration SQL scripts and managing schema evolution for batch execution metadata.
- Strengthened application security by remediating new and legacy vulnerabilities identified via **Seeker IAST** and **Checkmarx**, documenting standardized triage practices while maintaining **95%+ automated test coverage**.

- Owned a **stateless, asynchronous Product Orchestrator** built with **Spring Boot** and **Apache Camel**, coordinating requests across multiple downstream systems using **gRPC**, **REST**, and **SOAP** integrations.
- Implemented robust exception handling and **fail-fast semantics** in the orchestration layer to handle downstream unavailability, and validated resilience through targeted unit and route-level testing.
- Contributed to the **Product Gateway** by implementing **OAuth** flows including credential creation, refresh-token rotation, and secure token generation, along with data-mapping layers and gRPC models for downstream orchestration.
- Introduced automated testing practices using **JUnit** and **Mockito**, achieving **95%+ code coverage** across orchestrator routes and gateway services.
- Prototyped and introduced **feature-flagging** capabilities during internal hackathons, influencing the adoption of an in-house feature toggle mechanism within the Product Orchestrator platform.
- Evaluated observability, distributed tracing, and platform tooling including **OpenTelemetry** and **LaunchDarkly** through Proofs of Concept to inform architectural decisions.

EDUCATION

Kumaraguru College of Technology

Bachelor of Engineering in Electrical and Electronics Engineering

Aug 2016 – May 2020

Coimbatore, Tamil Nadu

PROJECTS

TXNomy | Python, Django, PostgreSQL (pgvector), PyTorch, React, Redis

Nov 2025

- Architected a **privacy-first AI transaction classification engine** achieving less than **34ms P50 latency** for semantic search using **PostgreSQL (pgvector)** and HNSW indexing.
- Engineered a **Hybrid Classifier Pipeline** combining **Fine-tuned mBERT (NER)** and **Semantic RAG** (Sentence Transformers) to outperform keyword baselines by **20% F1-Score** and achieve **100% transaction coverage**.
- Implemented a **Zero-Knowledge architecture** ensuring no user history persistence while enabling **community-sourced intelligence** via trust-weighted consensus algorithms.
- Built a high-throughput **Async Batch Processing system** using **Celery** and **Redis** to handle large-scale datasets with real-time SSE progress tracking.

Smart Aquarium Management System | Python, Flask, Tensorflow, Keras

Apr 2020

- Led the development of an automated Arduino-based IoT solution for Water Purity Maintenance in home aquariums and small-scale aqua-farms at less than half the cost of a Dissolved Oxygen (DO) sensor.
- Implemented a cloud-ready LSTM with an accuracy of over 90% to predict DO values from real-time aquarium data collected from pH, temperature, turbidity, and salinity sensors.

Automated Solid Waste Segregation | Python, Django, OpenCV, Tensorflow

May 2019

- Engineered an automated Solid Waste Segregation solution to accurately classify solid waste as Biodegradable or Non-Biodegradable with images from a PiCam coupled to a Raspberry Pi3.
- Leveraged OpenCV and TensorFlow CNNs for the Image Processing and Classification layer.

TECHNICAL SKILLS

Primary Languages: Java, TypeScript, Python, SQL

Backend & Integration: Spring Boot, Quarkus, Apache Camel, Kamelets, Camel K, gRPC, REST, SOAP, OAuth2, mTLS, Celery

Frontend: Angular, React, Server-side Pagination, Virtual Scrolling, Reactive Forms

Data & Storage: PostgreSQL, Redis, pgvector

AI / ML Systems: LangChain, Vector RAG, ChromaDB, PyTorch

Cloud & Infrastructure: Azure (APIM, ASB, Blob Storage), AWS, Kubernetes, Docker, GitHub Actions, Nginx, Linux

Observability & Quality: OpenTelemetry, JUnit, Mockito, Playwright, Checkmarx, Seeker IAST

Engineering Foundations: Distributed Systems, Microservices Architecture, Concurrency & Synchronization, Event-Driven Architecture, API Design, Fault Tolerance